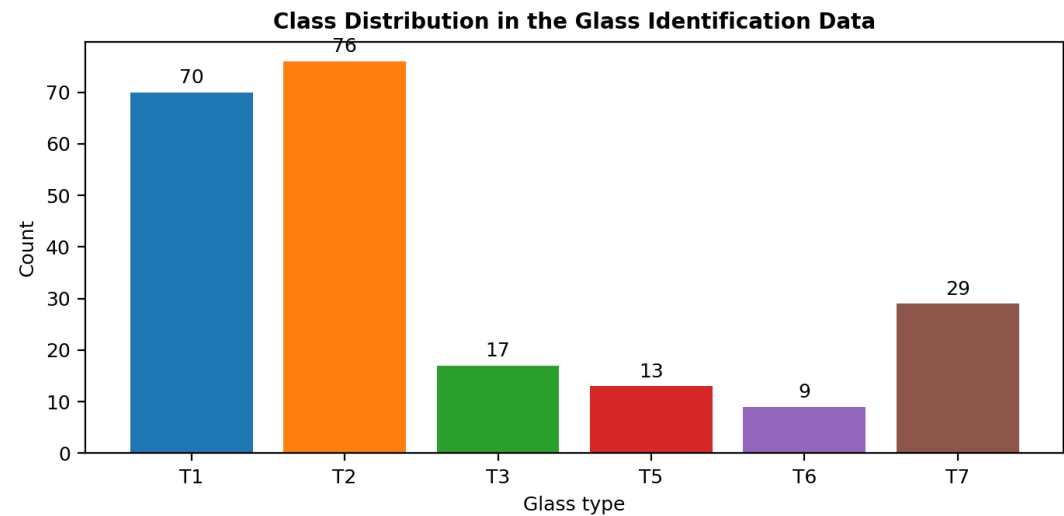


Glass Identification Dataset Description

Item	Details
Source	UCI Glass Identification Database; forensic chemistry data used for glass-type classification.
Files in the ZIP archive	glass.data, glass.names, glass.tag, and Index.
Observations	214 glass samples.
Variables	11 variables in total: 1 identifier, 9 continuous measurements, and 1 class label.
Observed classes	Six observed classes: Types 1, 2, 3, 5, 6, and 7 (Type 4 is absent).
Missing values	None.
Modeling variables	Eight variables were used in the report: RI, Na, Mg, Al, Si, K, Ca, and Ba.

The dataset was created for forensic science applications in which recovered glass fragments are classified by chemical composition. Each row is one glass specimen. The identifier is only a record number and is not analytically meaningful.

A small but important modeling choice was necessary for the multivariate report: Fe was excluded from the BACON and classification stages. The reason is technical, not cosmetic. The tableware class has only 9 observations, while the full chemical feature set has 9 quantitative variables. Standard within-class BACON requires $n > p$ so that covariance matrices are estimable in each class. Dropping Fe reduces p to 8 and makes the full sequence of within-class BACON, FLDA, and multinomial classification well-defined for all classes.



Class distribution for the observed glass types.

Class Labels

Code	Label	Count
1	Building windows (float)	70

2	Building windows (non-float)	76
3	Vehicle windows (float)	17
5	Containers	13
6	Tableware	9
7	Headlamps	29

Variable Dictionary

Variable	Meaning	Type	Unit	Role in report
Id	Record identifier	Identifier	-	No
RI	Refractive index	Continuous	Unitless	Yes
Na	Sodium	Continuous	Weight % oxide	Yes
Mg	Magnesium	Continuous	Weight % oxide	Yes
Al	Aluminum	Continuous	Weight % oxide	Yes
Si	Silicon	Continuous	Weight % oxide	Yes
K	Potassium	Continuous	Weight % oxide	Yes
Ca	Calcium	Continuous	Weight % oxide	Yes
Ba	Barium	Continuous	Weight % oxide	Yes
Fe	Iron	Continuous	Weight % oxide	No
Type	Glass type	Nominal class	Category	Response

Summary Statistics

Variable	Min	Max	Mean	SD
RI	1.511	1.534	1.518	0.003
Na	10.73	17.38	13.408	0.817
Mg	0.0	4.49	2.685	1.442
Al	0.29	3.5	1.445	0.499
Si	69.81	75.41	72.651	0.775
K	0.0	6.21	0.497	0.652
Ca	5.43	16.19	8.957	1.423
Ba	0.0	3.15	0.175	0.497
Fe	0.0	0.51	0.057	0.097

Fe has the smallest dispersion in the dataset ($SD \approx 0.097$), which is another reason it was the most defensible variable to omit when reducing the multivariate dimension from 9 to 8.

Analytical Notes

- Types 1, 2, and 3 are all window-glass categories and overlap substantially in chemical space; this overlap becomes the main source of classification error in the report.
- Types 5, 6, and 7 are smaller classes, but several of them are more chemically distinct, especially on Mg, Al, Ba, and K.
- Many variables are bounded at zero and display spikes or tail departures from normality, so perfect multivariate normality should not be expected.